
Day 3

Tasks

1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3	DDR4	DDR5
Typical Speed	800–2133 MT/s	1600–3200 MT/s	4800–8000+ MT/s
Voltage	~1.5 V	~1.2 V	~1.1 V
Performance	Basic/older performance	Faster and more efficient than DDR3	Fastest and most advanced of the three
Typical Use Cases	Older desktops, laptops and legacy systems	Modern/older gaming PCs, desktops and laptops	New-generation gaming PCs, workstations and high-performance desktops
Capacity	Generally lower	Higher than DDR3	Supports very high-capacity modules
Power Efficiency	Lower	Better than DDR3	Best power efficiency of the three
Compatibility	DDR3 motherboard only	DDR4 motherboard only	DDR5 motherboard only

Simple explanation

- **DDR3:** An older generation of RAM, commonly found in computers from the 2010s. It is suitable for basic computing and older systems.
- **DDR4:** A major improvement over DDR3, offering higher speeds and lower power consumption. It is widely used in many PCs and laptops from the last several years.
- **DDR5:** The newest generation, providing substantially higher memory speeds, greater capacity potential, and improved efficiency. It is commonly used in newer gaming PCs and high-performance systems.

Important: DDR3, DDR4, and DDR5 are not physically or electrically interchangeable. A DDR4 RAM stick, for example, cannot simply be installed into a DDR5 motherboard.

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2. Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.



➤ **Simple Steps to Remove and Reinstall RAM:**

1. Turn off the computer or laptop.
2. Unplug the power cable and charger.
3. Open the back cover (laptop) or side panel (desktop).
4. Find the RAM slot on the motherboard.
5. Push the two side clips outward. The RAM will pop up.
6. Remove the RAM carefully by holding the edges.
7. Take a photo of the RAM slot and RAM module.
8. Insert the RAM back into the same slot.
9. Press gently until the clips lock automatically.
10. Close the cover and reconnect the power.
11. Turn on the computer.
12. Check that the computer starts normally. If it boots successfully, the RAM is installed correctly.

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3. List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SSD (SATA)	NVMe SSD
Speed	Slowest; mechanical moving parts	Much faster than HDD	Fastest; uses PCIe interface
Durability	Less resistant to shocks because it has moving parts	More shock-resistant; no moving parts	More shock-resistant; no moving parts
Typical Use	Large, inexpensive storage and backups	Everyday computing, laptops and general applications	Gaming, video editing and high-performance applications
Daily app/use-case	Google Drive/backup storage – useful for storing large files locally at low cost	Microsoft Office – fast enough for everyday Word, Excel and PowerPoint use	Steam/PUBG/EA Sports FC – faster loading of games and large game files

In simple words

- HDD: Best when you need lots of storage at a low cost.
- SSD: Best for everyday computer use, quick booting and normal applications.
- NVMe: Best when you want maximum speed, especially for gaming, video editing and other demanding workloads.

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4. Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.
Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.

Practical Task

1. **Shut down** the laptop/PC completely and disconnect the charger/power cable.
2. Open the back/side panel according to the manufacturer's instructions.
3. Locate the storage drive:
 - **2.5-inch HDD/SSD:** usually uses a **SATA connector**.
 - **M.2 SSD:** a small stick-shaped drive installed directly into an M.2 slot.
 - **NVMe SSD:** usually an **M.2 NVMe drive**; NVMe describes the storage interface/protocol rather than the physical connector.
4. Carefully remove the drive.
5. **Take a clear photo of the connector/drive**, making sure the SATA or M.2 connection is visible.
6. Reinstall the drive securely.
7. Close the panel and reconnect the power.
8. Start the computer.
9. Confirm that the drive is recognized:
 - **Windows:** Win + X → Disk Management
 - Or open **File Explorer** → **This PC** and check whether the drive appears.
10. Take a screenshot/photo showing that the storage is recognized after reassembly.

What can I submit as the video link.

<https://www.youtube.com/watch?v=5F1WTh88OxA>
<https://youtu.be/5F1WTh88OxA?si=iphVgd0JP3nbva7w>